Lucas de Souza Brandão

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Education

Universidade de São Paulo, Bachelor of Computer Science — Feb 2022 – Present

- GPA: 9.1/10 (10th place among 92 students)

Skills

Technologies: Python (PyTorch, Pandas, NumPy, Scikit-Learn, Matplotlib, Librosa, HuggingFace, NetworkX).

Topics: Representation Learning, Music Information Retrieval, Computer Vision, Graph Neural Networks, Natural Language Processing, Speech Processing, Research, Machine Learning, Deep Learning, Data Science.

Languages: Portuguese (native), English (advanced) and Spanish (intermediate).

Research Experience

Full-Time Research Intern, **MTG**, UPF - Barcelona, Spain — May 2025 – Aug 2025

Personalized and interactive music retrieval based on similarity, using audio and text.

- Used **Audio-Language** models with **PyTorch** and **HuggingFace** to learn multimodal representations.
- Used **Essentia**, **PyFluidSynth**, **PrettyMIDI** and **Librosa** to process Audio and MIDI files.
- The work resulted in a paper under revision at **IEEE TASLPRO**.

Undergraduate Research, ICMC, USP — Oct 2024 – Present

Attribute-guided music retrieval using *Disentangled Representation Learning (DRL)*.

- Applied **Contrastive Learning** with **PyTorch** to generate audio *embeddings*.
- Developed a DRL method on **pre-trained foundation models**, resulting in a paper published at **LAMIR**.

Undergraduate Research, ICMC, USP — Aug 2023 – Jul 2024

Prediction of popular songs using Graph Neural Networks (**GNN**s), with **NetworkX** and **PyTorch Geometric**.

Undergraduate Research, ICMC, USP — Aug 2022 – Jul 2023

Evaluation of **Large Language Models (LLMs)** for the task of risk analysis.

- Used **LLM APIs**, **Pandas**, **NumPy**, **Scikit-Learn**, and **Matplotlib** to analyze data and structure the evaluation.

Projects & Leadership

RAIA — Aug 2024 – Sep 2026

- **Audio-MIDI Alignment:** — Aug 2025 – Sep 2026
  Research focused on measuring alignment between audio and MIDI pairs, under the supervision of senior researchers from **MTG**. The work resulted in a paper published at the **LAMIR** workshop.
- **TTS AutoTuning:** — Apr 2025 – Jul 2025
  Development of an automated *pipeline* for fine-tuning ***Text-To-Speech (TTS)*** models. Involved **fine-tuning** well-known pre-trained TTS models such as **XTTS-v2** and **Orpheus-TTS**. An automated dataset builder was developed from raw audio data using the **Whisper** model.
- **EcoVision:** — Aug 2024 – Dec 2024
  Development of a **Computer Vision** system to detect hazardous items in waste for the **Cooperativa Acácia**. We fine-tuned a **YOLO** model on our own custom dataset for **Oriented Bounding Box (OBB)** detection.

Data ICMC — Aug 2023 – Dec 2024

- Coordinated the **Data Science** and **Artificial Intelligence** group at ICMC-USP, leading its restructuring.
- Led a team in planning and developing courses on **Machine Learning**, **Data Science**, and **Python**, in addition to directly contributing to the preparation of the material and teaching the classes.